Ansible + Apache2 Setup Checklist

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Project: Configure two Debian machines and install Apache2 using Ansible.

Phase 1: Prepare both Debian machines

1. Machine 1 — Ansible Server

- [ ] Install Debian and boot the machine
- [ ] Update packages using apt update and apt upgrade
- [ ] Set hostname to ansible-server
- [ ] Identify and record the server IP address
- [ ] Install Ansible, OpenSSH client, and Python 3
- [ ] Verify Ansible, SSH, and Python versions
- [ ] Create the ~/ansible-lab working directory

2. Machine 2 — Managed Node

- [ ] Install Debian and boot the machine
- [ ] Update packages using apt update and apt upgrade
- [ ] Set hostname to ansible-node1
- [ ] Identify and record the node IP address
- [ ] Install openssh-server, Python 3, sudo, and curl
- [ ] Enable and start the SSH service
- [ ] Create the dedicated ansible user
- [ ] Add the ansible user to the sudo group

3. Network and SSH Configuration

- [ ] Verify both machines can communicate over the network
- [ ] Confirm TCP port 22 is reachable from Machine 1
- [ ] Generate an Ed25519 SSH key on Machine 1
- [ ] Copy the public key to Machine 2
- [ ] Test SSH login from Machine 1 to Machine 2
- [ ] Verify the ansible user can execute sudo commands

4. Ansible Configuration

- [ ] Create inventory

- [ ] Add the managed node IP and ansible_user to the inventory
- [ ] Set the Python interpreter path in the inventory
- [ ] Verify the inventory using ansible-inventory
- [ ] Run the Ansible ping module and confirm pong
- [ ] Test privilege escalation using become and -K

5. Apache2 Installation

- [ ] Create the install-apache.yml playbook
- [ ] Add the webservers target group
- [ ] Configure the playbook to use become: true
- [ ] Add a task to update the APT cache
- [ ] Add a task to install Apache2
- [ ] Add a task to enable and start Apache2
- [ ] Add a task to verify Apache2 service status
- [ ] Run the playbook syntax check
- [ ] Execute the playbook and confirm zero failures

6. Final Verification

- [ ] Verify Apache2 is installed on Machine 2
- [ ] Verify Apache2 is enabled at boot
- [ ] Verify Apache2 service is active
- [ ] Test HTTP access using curl from Machine 1
- [ ] Open the managed node IP in a web browser
- [ ] Confirm the default Apache page loads
- [ ] Run the playbook again and check idempotency

Expected final result: Ansible Server successfully manages the Debian node over SSH, Apache2 is installed and running, and the website responds over HTTP.